Horizon I and II OPO Operation and Maintenance Manual

Continuum®

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P/N: 996-0034, Revision D

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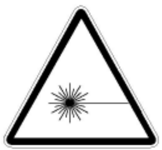
Preface

The internationally recognized warning symbols shown here are used throughout the manual to alert the user of potential hazards when operating this laser.



LASER RADIATION:
The radiation symbol is to alert the user to the danger of laser radiation when performing certain operations and when the danger of exposure is the greatest. It is also used when the optical hazards are present.

DANGER VOLTAGE:
The lightning or thunderbolt indicates the presence of high voltage which may pose a danger to the user or the equipment.

CAUTION:
The exclamation point symbol is to draw attention to other potential hazards not included under the optical or electrical categories. The user should be aware of special care that needs to be taken when performing potentially hazardous procedures.

WARNING: PERSONAL INJURY
The Safety section is placed as the first chapter of the manual to emphasize its importance and the need for everyone to read it before working with the laser.

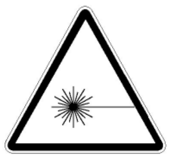
The installation section includes a copy of the installation instructions for future reference. The site requirements are listed, as is the initial installation process step by step.

The Troubleshooting chapter presents common problems that are encountered and ways to solve them in both the electronic and optical areas. It is here that the more complicated procedures such as eliminating free-running are discussed.

The Appendix lists the Continuum warranty and gives Service telephone and FAX numbers for further help. The Operator Notes gives the user a convenient place to keep maintenance records and other information.

Préface

Les symboles de danger reconnus internationallement présentés ici sont utilisés tout au long du manuel pour avertir l'utilisateur des risques possibles lors de l'utilisation de ce laser.



Le symbole de radiation sert à avertir l'utilisateur du danger d'exposition à la radiation laser lors de l'exécution de certaines opérations et lorsque ce danger est le plus grand. Il est aussi utilisé en cas de risques optiques.

Le signe d'éclair indique la présence d'une haute tension qui peut représenter un danger pour l'utilisateur ou l'équipement.

Le point d'exclamation sert à attirer l'attention sur d'autres risques possibles non inclus dans les catégories optique et électrique. L'utilisateur devrait être conscient des mesures de protection particulières à prendre lors de l'exécution de procédures potentiellement dangereuses.

La section Sécurité est en tête du premier chapitre de ce manuel afin de souligner son importance et la nécessité incombant à chaque utilisateur de la lire avant de commencer à utiliser un laser.

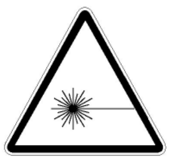
La section Installation comprend une copie des instructions de montage pour future référence. Les exigences de site sont énumérées comme l'est le processus d'installation initial, étape par étape.

Le chapitre Dépannage présente les problèmes couramment rencontrés et indique les mesures à prendre dans les domaines de l'électronique et de l'optique. C'est aussi dans ce chapitre que les procédures plus complexes telles que l'élimination du fonctionnement libre sont expliquées.

L'Annexe décrit la garantie Continuum et fournit les numéros de téléphone et de fax du service après-vente. Les Notes de l'Utilisateur offrent à l'utilisateur un endroit pratique pour conserver les rapports d'entretien et d'autres informations utiles.

Vorwort

Die hier abgebildeten, international verständlichen Warnsymbole werden in der gesamten Anleitung dazu verwendet, den Benutzer auf mögliche Gefahrenquellen beim Laserbetrieb hinzuweisen.



Das Strahlungssymbol warnt den Benutzer vor gefährlicher Laserstrahlung bei bestimmten Betriebsverfahren und bei den stärksten Strahlengefährdungen. Es wird auch für optische Gefahrenquellen verwendet.

Das Blitzsymbol weist auf das Vorliegen von Hochspannung hin, die eine Gefährdung für Benutzer oder Gerät darstellt.

Das Ausrufezeichensymbol soll auf weitere Gefahren aufmerksam machen, die nicht unter die Kategorie Optik oder Elektrik fallen. Der Benutzer sollte auf besondere Maßnahmen achten, die bei evtl. gefährlichen Verfahren anzuwenden sind.

Das erste Kapitel dieser Anleitung ist der Sicherheitsabschnitt, da dieser äußerst wichtig ist und vor dem Betreiben des Lasers von jedem Benutzer gelesen werden sollte.

Der Abschnitt zur Installation enthält für den zukünftigen Bedarf ein Exemplar der Installationsanleitung. Hier sind sowohl die Standortanforderungen als auch eine schrittweise Erläuterung der Erstinstallation enthalten.

Das Kapitel Fehlersuche und -behebung beschreibt bekannte Probleme (Elektronik und Optik), die auftreten können, sowie Abhilfemaßnahmen. Hier werden auch kompliziertere Verfahren, wie das Verhindern von “Freischwingern”, erörtert.

Der Anhang enthält die Continuum-Garantie sowie Service-Rufnummern und -Faxnummern, unter denen weitere Hilfestellung verfügbar ist. Unter “Benutzernotizen” kann der Benutzer bequem Wartungsdaten und sonstige Informationen aufbewahren.

Chapter I Laser Safety Precautions

A. Class IV laser safety precautions

Continuum's Horizon OPO is a Class IV high power coherent light source whose beams are, by definition, a safety hazard. Take precautions to prevent accidental exposure to both direct and indirect (reflected) beams. Diffuse as well as specular beam reflections can cause severe eye and skin damage.

The Horizon OPO pump beam and some Horizon OPO outputs are invisible, making them even more dangerous. Infrared and ultraviolet radiation passes easily through the cornea, which focuses it onto the retina where it can cause instantaneous permanent damage or blindness. Even small doses from scatter can be harmful.

B. General safety rules

Warning!

Avoid eye and skin exposure to direct or indirect radiation! If the equipment is used in a manner not specified by the manufacturer, the protection provided by the equipment may be impaired!

The following is a summary of general safety precautions, which are to be observed by anyone working with the Horizon OPO. For a complete listing of safety standards, see the ANSI booklet listed under Government and Industry Safety Regulations; see later in this Chapter.

Keep all unnecessary personnel out of the work area. Remove all shiny reflective surfaces (including rings, watch bands, metal pens, etc.) and all flammable materials with their containers from the work area.

Operate lasers only in well-marked areas with controlled access. Be sure to post appropriate warning signs visible to all.

Operate lasers under the supervision of qualified personnel only. When not in use, shut down the laser completely.

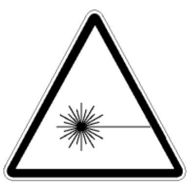
Provide interlocks for all doors.

A fire resistant background should be in place behind the target area. The surrounding areas should be coated with a material that absorbs any scattered radiation that might occur.

C. Electrical safety rules

Turn off all power to the pump laser before beginning maintenance or repair of the Horizon OPO or pump laser. Electrical shock and burns resulting from input power or capacitor discharge can cause serious injury or death.

D. Optical safety rules



Eye safety is a great concern. The Horizon OPO and its pump laser are Class IV light sources, the most dangerous classification. Even a reflection can cause permanent, severe eye damage. Never look at a beam or reflection. Specular reflections from the main beam off a polished surface can cause severe eye damage.

Everyone in sight of the Horizon OPO must always wear laser goggles. The goggles must block the invisible 355 nm pump beam; even small doses of its scatter are harmful. Goggles must also have the appropriate protection value for the output wavelengths produced, 192-2750nm, and must meet current ANSI Z136.1 standards, see later in this Chapter.

Wear goggles while adjusting the Horizon OPO, even when the pump laser beam is attenuated. Dangerous scatter from the beam dumps and spurious reflections can move while optics rotate.

Do not wear or use any object that may reflect light such as a watch, ring, pen, chromed tool, etc.

Light the area around the laser so that the operator's pupils are constricted normally.

ONLY operate the Horizon OPO when all the covers are in place.

Close beam exit shutters when the Horizon OPO is not in use.

Use an IR detector or energy detector to verify that the beam is off before working in front of the Horizon OPO.

Set up experiments so beams are not at eye level.

Avoid vertical beam paths; shield them completely if you must use them.

Provide enclosures for beam paths whenever possible.

Avoid blocking any beam or reflection with any part of your body.

Mark the lab well with warning signs when the laser is operating and provide interlocks for all doors.

E. Safety features

Covers and interlock

The Horizon OPO cover is interlocked for your protection. Once this cover is opened the interlock is activated to immediately stop the system from lasing operation. If you choose to defeat this interlock for fine adjustments or servicing the Horizon OPO, then you need to be aware of the possible dangers – particularly from stray reflections.

The interlock feature only works if both the Nd:YAG and OPO are "daisy-chained" together. Be sure that this connection has been made – locate the simple molex connector beneath the Horizon bench and connect this to the Nd:YAG Interlock connector. We strongly advise using this feature for your overall safety.

Safety labels

Safety labels attached to the Horizon OPO warn of specific dangers. **Figure 1.1** shows the location of Horizon OPO safety and warning labels. This is a Class IV system – take time to be familiar with the layout of the laser and the level of emission from each exit port.

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<tr>
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Item 1
P/N 310-0029

Found on internal covers

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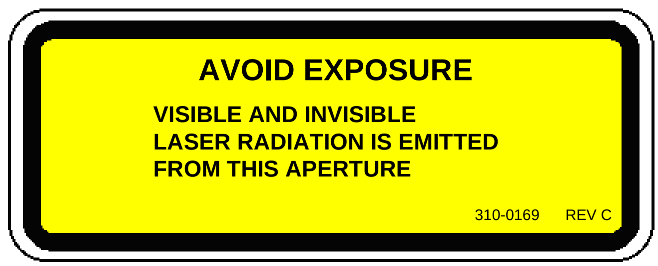
Item 2
P/N 310-0169



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Item 3
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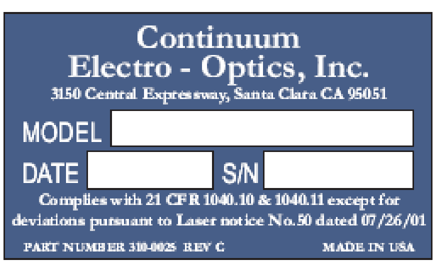

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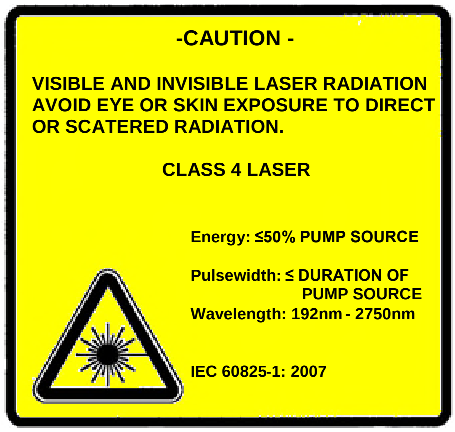
-CAUTION -
VISIBLE AND INVISIBLE LASER RADIATION
AVOID EYE OR SKIN EXPOSURE TO DIRECT
OR SCATERED RADIATION.
CLASS 4 LASER
Energy: ≤50% PUMP SOURCE
Pulsewidth: ≤ DURATION OF
PUMP SOURCE
Wavelength: 192nm - 2750nm
IEC 60825-1: 2007


Horizon
USB
12V.DC.10A


Continuum
Electro - Optics, Inc.
MODEL
DATE
S/N
CE
This device complies with part 15 of the FCC Rules.
Operation is subject to the following two conditions:
(1) this device may not cause harmful interference, and
(2) this device must accept any interference received
including interference that may cause undesired
operation.
This Class A digital apparatus complies with Canadian
ICES-003
Cet appareil numerique de la classe A est conforme
a la norme NMB-003 du Canada.
MADE IN USA
310-0309 REV B

F. Government and industry safety regulations

Continuum strongly suggests that all its customers purchase a copy of the **American National Standard for the Safe Use of Lasers**(ANSI Z136.1) in order to read and implement necessary precautions. The American National Standards Institute (ANSI), a member of the International Organization for Standardization (IOS) and the International Electrotechnical Commission (IEC), issues this booklet. Write or call the publisher listed below to obtain a copy.

Continuum's user information complies with section 1040.10 of 21 CFR Chapter I, Subchapter J concerning Radiological Health, published by the U.S. Department of Health & Human Services, Center for Devices & Radiological Health

Laser Institute of America
13501 Ingenuity Drive, Suite 128
Orlando, FL 32826
USA

Telephone:	+1 (800) 345-2737
FAX:	+1 (407) 380-5588
E-mail:	lia@laserinstitute.org
Web:	www.lasertitute.org

G. Additional safety references

Regulations of the Administration and Enforcement of the Radiation Control for Health and Safety Act, 1968. US Dept. of Health and Human Services, Public Health Service and FDA

A Guide for Control of Laser Hazards. American Conference of Governmental Industrial Hygienists

European Committee for Electrotechnical Standardization:

EN 61010-1

EN 60825-1

Declaration of Conformity

We
Continuum Electro-Optics Inc.
140 Baytech Drive, San Jose, CA 95134, USA

Hereby declares that under our sole responsibility the components/products:

Optical Parametric Oscillator (OPO)

Models: Horizon I and Horizon II

being optional accessories to our Surelite and Powerlite DLS systems are in conformity with the provisions of the following EC Directives, including all amendments, and national legislation implementing these directives:

- 2004/108/EC Electromagnetic Compatibility Directive (EMC)
- 2006/95/EC Low Voltage Directive (LVD)

And that the following harmonized standards have been applied

EN61010-1: 2010
EN60825-1: 2007
EN61326-1: 2006 Class A
EN55011/CISPR 11
EN61000-3-2: 2006
EN61000-3-3: 2008
EN/IEC 61000-4-2, EN/IEC 61000-4-3,
EN/IEC 61000-4-4, EN/IEC 61000-4-5,
EN/IEC 61000-4-6, EN/IEC 61000-4-11

Place of Issue: Santa Clara, CA
Date: December, 2012

Larry Cramer
President & CEO

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Chapter II Installation

Continuum will perform the initial Horizon OPO installation and train its operators. The information in this chapter will assist in the initial installation as well as subsequent ones. It is recommended that the user contact Continuum Service before moving or reinstalling the Horizon OPO.

Tools and supplies

Please have these supplies on hand for the Horizon OPO installation; they are also necessary for subsequent alignment and maintenance. Also refer to ***Horizon OPO Site Requirements*** later in this Chapter. The welcoming letter sent on receipt of your order also lists these items.

- Laser goggles (1 pair per person in the room)
- Finger cots or fine cotton or surgical gloves
- High-intensity flashlight
- Dental inspection mirror
- Eyedropper, lint-free lens tissues and cotton swabs
- Source of clean, dry gas under pressure (not Freon-based)
- Flat-blade screwdriver and hemostats (surgical pliers)
- Dry, spectrophotometric grade acetone, 100 ml
- Dry, spectrophotometric grade methanol , 100 ml
- Distilled water, 3 gallons
- Metric Allen wrench set

- Red phosphor viewing card (Kodak)
- White viewing card (business card or other bleached paper)
- Burn paper (e.g., Kentek Zap-It alignment paper, http://www.kenteklaserstore.com)
- Sealable plastic bags to hold burn paper, about 2” x 3.5”
- Power meter with 10-30 watt absorbing head, calibrated from 200-2750 nm, time constant <4 sec, sensitive to 10 Hz, 1-100 mJ, 2-8 nsec pulses
- Fast (<0.5 nsec) photodiode (Electro-optics ET2000)
- Oscilloscope (350 MHz) or digital scope (5 GHz)
- Etalon(s) for working range(s), 10 GHz free spectral range, Finesse >30
- Linewidth analyzer or pulsed spectrum analyzer (e.g., Burleigh PLSA 3500).
- 100-240VAC, 47-63Hz, 1.58-0.6A – required to power the external 12 VDC, 10A power supply included with the Horizon OPO system.

Installation overview

Follow the instructions given in this chapter to prepare the Horizon OPO for first-time installation, but **do not** attempt to install the system.

Note:

Do not connect any cables to the Horizon OPO before the Continuum service engineer arrives or the warranty may be voided.

Please perform *only these steps* listed below in advance of the Continuum customer Service Engineer's arrival:

1) **Schedule the Continuum customer service engineer** for your installation when the equipment arrives (call 1-877-272-7783).

2) **Prepare the site** for the equipment, following the safety rules given in **Chapter I, *Laser Safety Precautions***, and in ***Horizon OPO Site Requirements*** (later in this Chapter).

3) **Inspect the crate upon arrival** for shipping damage and notify the shipper if any damage is visible. Look to see if the Shock Meter has been tripped. If so, please report to Continuum Service.

4) **Unpack the system** and check that all parts are present as listed in the packing list. The Horizon OPO is shipped in one crate along with the GUI software, a 12V power supply and accessories. Lifting the Horizon requires two people lifting the unit using the openings located at the bottom of the short sides.

Note:

Notify Customer Service if any parts are missing.

5) **Check all parts for shipping damage.** Any shipping damage is the responsibility of the shipping company and the buyer. Open the Horizon OPO cover and check for loose parts. Make a note of anything that seems to be out of place and if necessary report to Continuum Service.

Note:

Notify both the shipper and Continuum Customer Service of any damage immediately.

6) A Continuum Service Engineer will complete the installation by aligning the Horizon OPO and pump laser, installing the Horizon software onto the computer, demonstrating the Horizon OPO specifications, and training its operators.

Horizon OPO site requirements

Warning!

Follow all Chapter I safety regulations in choosing a location for the Horizon OPO.

Room air

Locate the Horizon OPO in a clean area, where the temperature is stable during the working day. If the room is air conditioned, or if air filters are present, place the equipment away from the moving air.

Operating temperature range should be kept between 18°C to 30°C (65° F to 87° F) within ±3°C (5° F). Keep humidity below 50%. A laminar flow air filter over the Horizon OPO can significantly reduce the amount of dust in the air. This will significantly reduce the chance of dust depositing on optical surfaces that could lead to damage.
Non-Operating temperature range: 10°C to 39°C (50° F to 102° F).
Non-Operating Humidity range: 5-95% non-condensing.

Optical bench

Use a stable optical table large enough for the Horizon OPO, pump laser and all other accessory equipment; an NRC table, tapped with a maximum of two inch centers is ideal. Allow access from all sides. Ground the pump laser and Horizon OPO well; check with an oscilloscope. If vibration sources are nearby (e.g., a vacuum pump), use vibration-damping table supports.

Pump laser

The pump laser beam quality is critical to the performance of any Optical Parametric Oscillator. We strongly suggest using one of Continuum's Nd:YAG lasers for optimal performance from the Horizon OPO. Continuum fully characterizes the pump laser prior to pumping an OPO to ensure optimal OPO performance and to eliminate any irregularities in the pump beam that could cause expensive crystal damage.

If pumping with the Continuum's Powerlite DLS lasers, you need to provide 208 VAC, single-phase, 30 Amp of power, and a constant pressurized water source (<65°F). If pumping with the Continuum's Surelite lasers, provide 208 VAC, single-phase, 10 Amp of power. Please refer to your Continuum pump laser manual for more detail.

Placement

Ensure that both the Nd:YAG pump lasers and the Horizon OPO are level and adjusted to match the correct input beam height. Firmly attach both systems to an optical table with the foot clamps supplied.

The pump laser should be configured to provide the beam quality needed to pump an OPO.

Horizon Energy Specifications

Tuning Range* (nm)		400-2750	192-400
Pump Laser		Energy at Peak (mJ) 425nm	Energy at Peak (mJ) 325nm
Pump Model	OPO Pump Energy (mJ) at 355nm		
Horizon I pumped with 200mJ or less			
SL I-10	100	30	6
SL II-10	160	50	10
Horizon II pumped with 200mJ or more			
SL EX	220	80	15
PL 8000	290	110	20

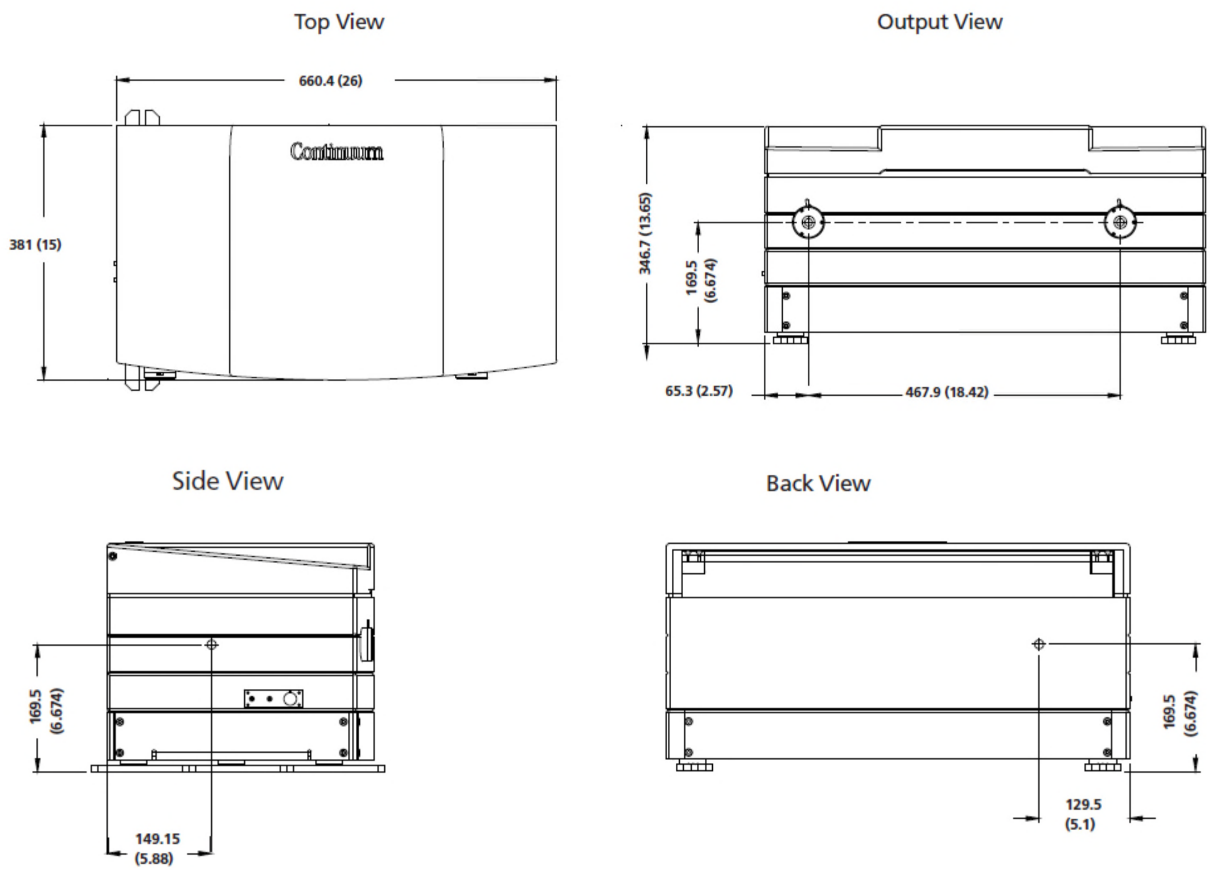
Horizon Specifications

Description	SL I, II, EX	PL 8000
Pulsewidth (nsec)	3-5	3-7
Pointing Stability (µrad)	<±100	<±100
Linewidth (cm^{-1})		
Seeded	2-6	
Unseeded	3-7	
Energy Stability (%, 99% of shots)	<±10	
Divergence (mrad, FWHM)	<2 (both axes)	
Beam Diameter (mm, near field)	4-7	
Beam Roundness (%, near field)	>85	
Polarization (%)		
Signal Horizontal	>99	
Idler Horizontal	>99	

Figure 2.2
External dimensions of the Horizon I and II OPO

Horizon Package Dimensions

Size	LxWxH	660.4 x 381 x 346.7 mm (26 x 15 x 12.33”) H ±12 mm/0.5”
Weight		30.8 kg (68 lbs)



Connecting the safety interlock

The Horizon OPO and Continuum Nd:YAG laser are interlocked with a 3-prong Molex connector for your safety. When the connector's two outer contacts are shorted, the interlock is defeated.

The safety interlock will shut off the pump laser when the Horizon OPO lid is lifted, or when any other piece of equipment (e.g., the pump laser cover or a door switch) that is daisy-chained to this interlock opens the interlock circuit.

Warning!

Operation with the safety interlock defeated is extremely dangerous. Observe all safety procedures scrupulously while the cover is open.

To link the Horizon OPO safety interlock with a Continuum pump laser, remove the jumper plug from its three pin connector under the Nd:YAG chassis near the output end of the laser and connect it to the matching three-prong connector of the Horizon OPO. Retain the jumper plug for future use.

Software installation

A. Hardware requirements

Installation

Outlined are the requirements (hardware and software) and procedures pertaining to the installation, launch, and removal of the application software. Please see the Horizon GUI manual for a full explanation of the operating platform for the Horizon OPO.

Hardware minimum requirements

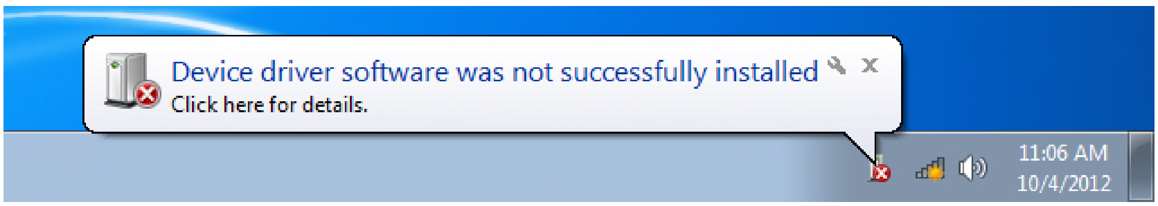
- IBM PC Compatible computer with 2.8 GHz Dual Core processor
- 1GB free hard disk space
- 2GB RAM
- USB 2.0 (Additional USB port + USB to serial converter or Serial port to DLS laser)
- 24X CD-ROM drive.

B. Software requirements

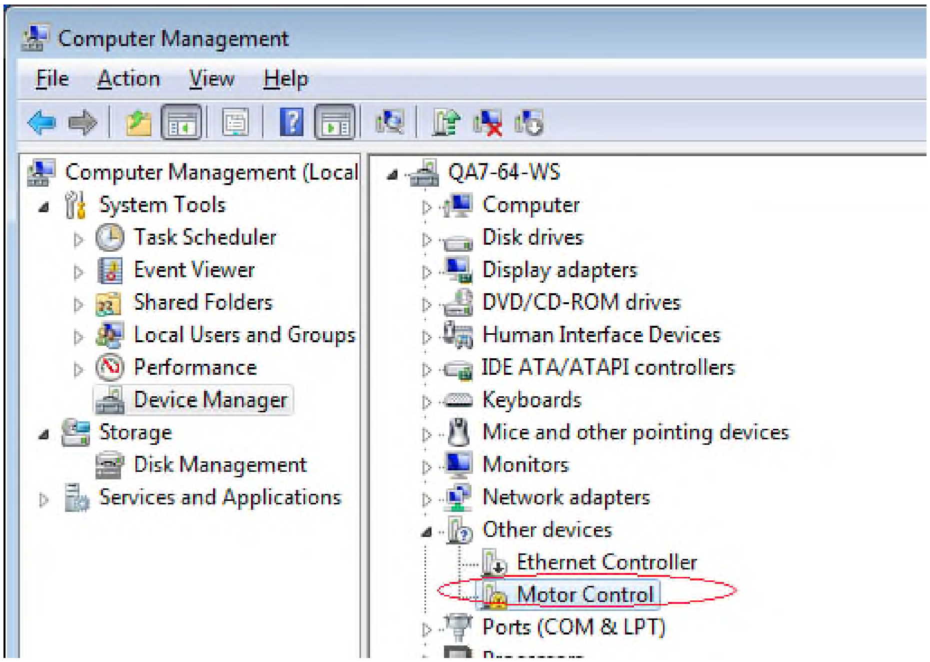
- Windows XP or Windows 7 (32 bit or 64 bit). The application has been tested on Windows XP Professional and Windows 7.
- The .NET Framework version 4.0 redistributable package. This is a free application available for download from Microsoft®. http://www.microsoft.com/downloads

C. Installation procedure (motor control driver)

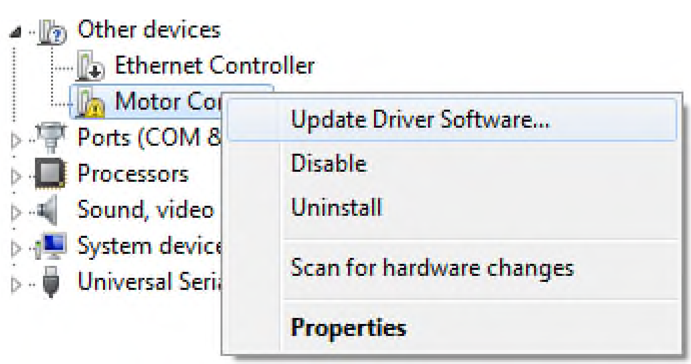
- Insert the CD into the CD-ROM drive. For example, assume that the CD drive is the D: drive.
- Prior to commencing installation, it is recommended that all open applications are closed and the .NET Framework has been installed.
- Since the communication is USB, the motor control driver needs to be installed.
- Relevant USB driver under the "Prerequisites/Motor Control Driver Horizon" folder.
- Power up Horizon hardware
- Connect the Horizon USB cable to the PC.
- When the USB is plugged on to the PC it will recognize the USB hardware and look for drivers.
- Then following message will appear.



- Go to the Device Manager window and locate the Motor Control as shown below.

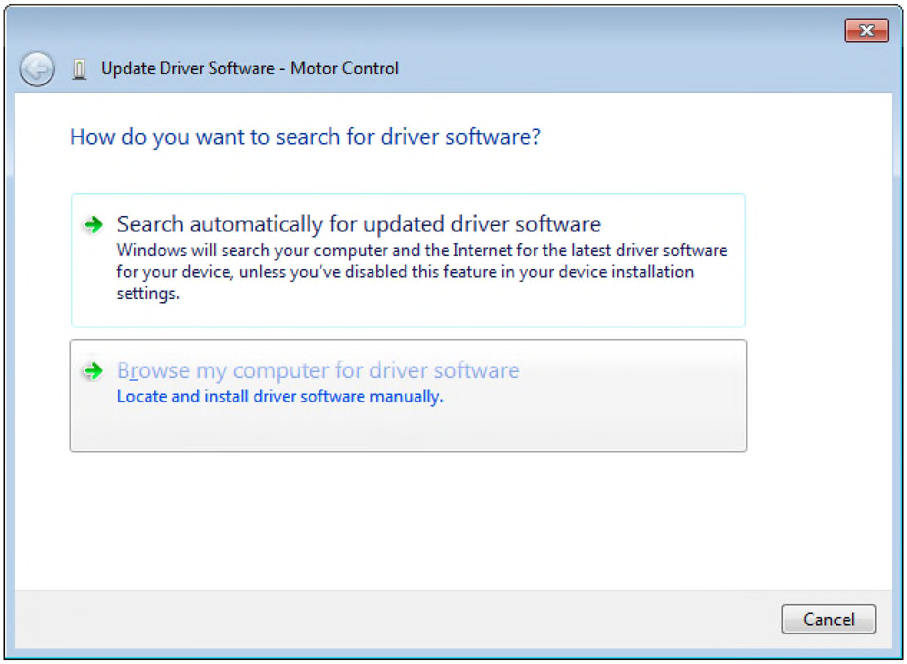


Then right click on the Motor Control and Select "Update Driver Software" Menu Item.

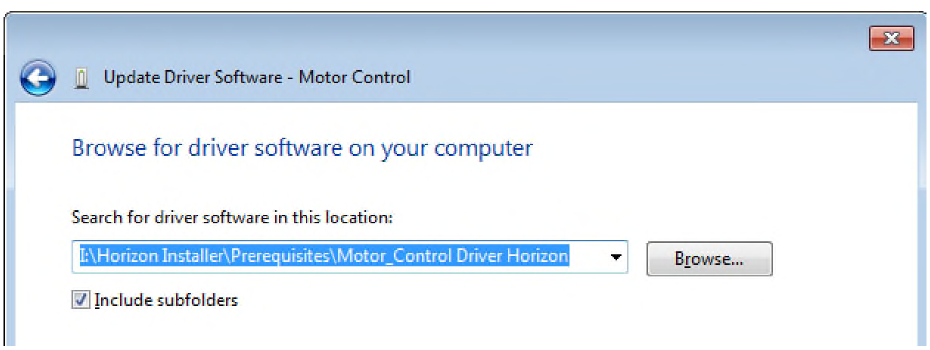


The following window will appear.

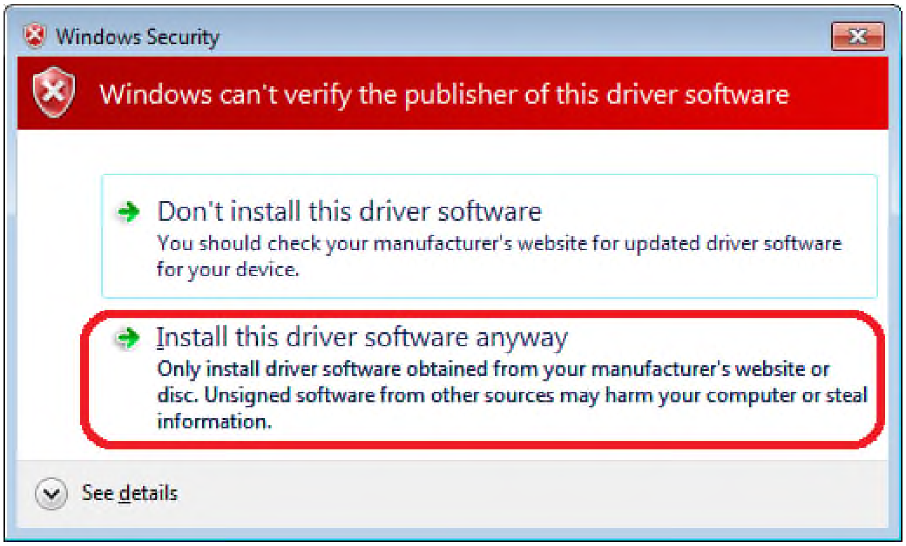
Select the option “Browse my computer for driver software”.



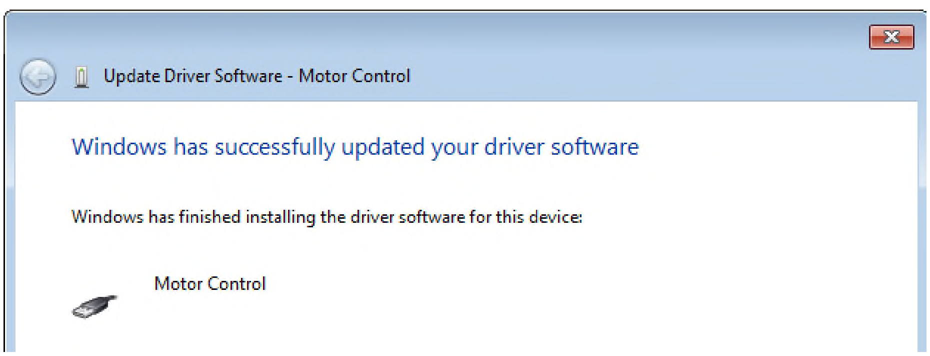
- Locate the motor control driver from the CD to install the driver as shown in the window below.



Please select “Install driver software anyway” from the following window.



- Once you successfully install the driver, windows will let you know that the driver installation has been successful.



D. Software installation

- Open Windows Explorer by Right-clicking the Start button and clicking Explore, or any other technique that you are familiar with.
- Double click on the D: drive (the CD ROM drive is assumed to be D:)
- Double click on the D:\ Prerequisites\
- Double click on LibUsbDotNet_Setup.2.2.6.exe
- Follow the instructions in the setup program.
- Once the above installation is complete, double click on setup.exe at D: drive.
- Follow the instructions in the setup program.

- Remove the CD-ROM and store it in a safe place. The CD is not needed to run the software after the installation is complete.

It is advisable to not have other software installed on the PC that will be driving the Horizon GUI. This will minimize the possibility of software conflicts.

Chapter III System Operation

A. Daily operation

Quick start

1) Ensure that the pump laser shutter is closed for your safety. Set the pump laser in flash mode for a 15-20 minute warm up period – if operating in seeder mode the seeder should be switched on for approximately 45mins to guarantee a stable output. (We suggest leaving the seeder switched on at all times to eliminate this 45 minute warm up period each morning). Open the shutter and activate the Q-switch when ready.

Warning!

Wear goggles and observe all safety precautions.

Caution!

Always check the pump beam energy and seeding (if applicable) before pumping the Horizon OPO.

2) Verify that the Horizon's 12VDC power supply is connected to the Horizon, plugged into AC power and has been running for a minimum of 10 minutes to allow the OPO and UV crystal heaters to stabilize. Again, this can be left on overnight so eliminating a cold start each day.

3) Open the Horizon GUI software. Set to the operating wavelength and run your system.

Shutdown

1) Follow the standard shut down procedure for your Nd:YAG laser. Refer to Continuum's manual for the Surelite and/or Powerlite as needed.

2) Deactivate the Nd:YAG Q-switch, stop flashing mode and close the shutter.

3) Close all Horizon OPO output shutters.

4) Exit the Horizon GUI software – as stated in the GUI manual we advise selecting ***Exit*** from the Tools menu bar or closing the GUI window directly. These procedures will ensure that the crystals "Park" safely at the end of the day – eliminating the possibility of collision at start up each day.

Caution!

Humidity in the crystal ovens can damage hygroscopic BBO crystals. Continuum recommends that the 12VDC external power supply be kept on at all times to protect the crystals.

Complete Start Up

1) Ensure that the pump laser shutter is closed for your safety. Set the pump laser in flash mode for a 15-20 minute warm up period – if operating in seeder mode the seeder should be switched on for approximately 45mins to guarantee a stable output. (We suggest leaving the seeder switched on at all times to eliminate this 45 minute warm up period each morning). Open the shutter and activate the Q-switch when ready.

Warning!

Wear goggles and observe all safety precautions.

Caution!

Always check the pump beam energy and seeding (if applicable) before pumping the Horizon OPO.

2) Verify that the Horizon's 12VDC power supply is connected to the Horizon, plugged into AC power and has been running for a minimum of 10 minutes to allow the OPO and UV crystal heaters to stabilize. Again, this can be left on overnight so eliminating a cold start each day.

3) Open the Horizon GUI software and GoTo 508nm. This is the peak of the OPO signal output. At this wavelength the crystals and Pellin Broca are set at optimal positions for maximum output from the Horizon.

4) Open the Horizon OPO top cover. The interlock will be activated – this needs to be defeated to follow the active alignment through your OPO. Be sure to take all precautions at this time – especially be aware of any stray reflections within the Horizon OPO.

Warning!

Wear goggles and observe all safety precautions.

5) Check the OPO pump beam alignment - reduce the 355nm input power to approximately 1mJ by delaying the Q-switch and follow the beam path through the initial alignment path. Note that the input beam and retro-reflection from the OPO should pass through a pinhole jig placed at location A. The pump beam and its reflections should be centered on pinholes A.

6) Block the pump beam and insert tool T1 between location C and D. Remove the plug on the side exit from the Horizon. Adjust T1 to steer the pump beam through this exit. Place a power meter to monitor output energy.

7) Unblock the pump beam and measure pump energy. Set Q-delay of the pump laser (if applicable) to original setting to bring the OPO pump level up to full power (see Horizon specifications for the maximum pump levels suggested depending on your system configuration). The pump laser should propagate with little divergence and again show no hot spots in the beam burns at full power.

8) With a fast photodiode check pump laser seeding if applicable. It is advisable to let the system run for a few pulses to ensure that the seeding is constant prior to pumping the OPO.

9) Block the pump beam and remove tool T1. Insert the plug to close the side exit port of the Horizon. Open the 355 nm pump laser beam dump and admit the pump beam.

10) If the energy is not adequate, follow the procedure in Chapter 3 Horizon OPO Oscillator calibration. It can be calibrated to the wavelength of your interest.

11) Replace the cover on the Horizon OPO. The system should be fully operational and optimized at this point.

UV Phase matching (for slightly low UV power)

1) Block the pump beam into the Horizon, or deactivate the Q-switch and close the pump laser shutter.

2) Follow the UV calibration procedure in Section “G” to optimize the crystal orientation and enhanced UV output.

Warning!

Wear goggles and observe all safety precautions.

3) Your system should now be optimized.

Table 3.1

Legend to Horizon OPO Layout		
Item #	**Part/Assy #**	**Description**
1, 3, T1	105-0023	Mirror dichroics, 355 nm
2a	108-0002	HALF WAVE PLATE *(option)*
2b	199-0129	POLARIZER 355 nm (*option*)
4	101-0214	LENS
5	102-0095	LENS
6	110-0007	WINDOW
7	109-0067	PORRO PRISM
8	108-0041	HALF WAVE PLATE
9	105-0294	INJECTION PUMP MIRROR
10	202-0270	# 1 Crystal BBO Type-1
11	202-0269	# 2 Crystal BBO Type-1
12	105-0293	MIRROR
13	110-0107	WINDOW
14 (A)	109-0069	POLARIZER *(option)*
15 (B)	108-0041	HALF WAVE PLATE
16	110-0108	WINDOW
17	202-0264	CRYSTAL, BBO
18 (C)	108-0042	WAVEPLATE
19	105-0292	MIRROR
20	202-0263	CRYSTAL, BBO
21	110-0070	WINDOW
22	109-0068	PELIN BROCA
23	105-0295	MIRROR
24, 29	105-0002	MIRROR
25a	108-0004	HALF WAVE PLATE ***(PL9000 only)***
25b	199-0055	POLARIZER ***(PL9000 only)***
30	108-0004	HALF WAVE PLATE
26 (D)	619-0720	BEAM DUMP ASSY ***(PL9000 only)***
35	109-0070	PELIN BROCA
38, 39	109-0008	PRISM 90 DEG

Figure 3.1
Horizon OPO optical layout

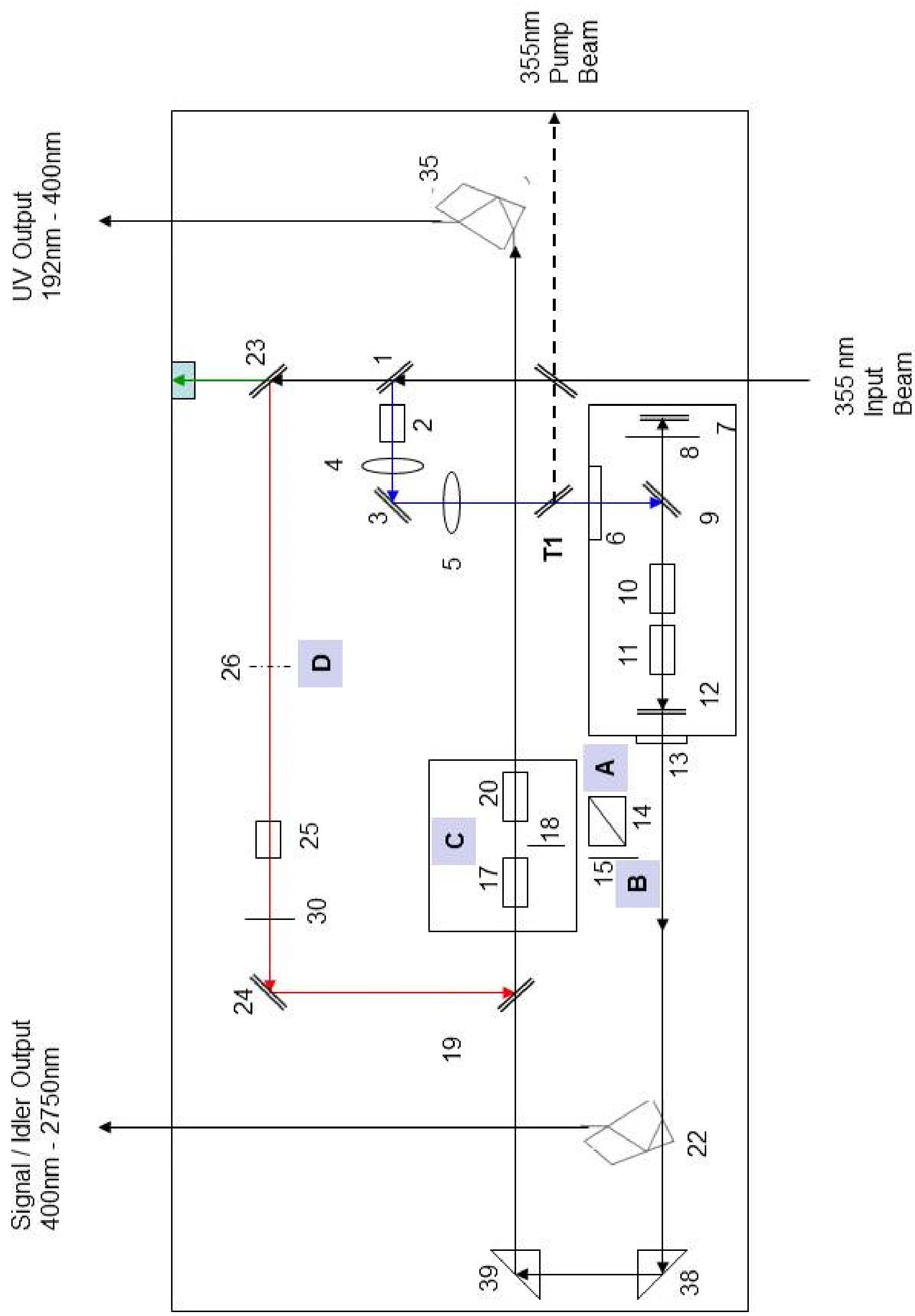
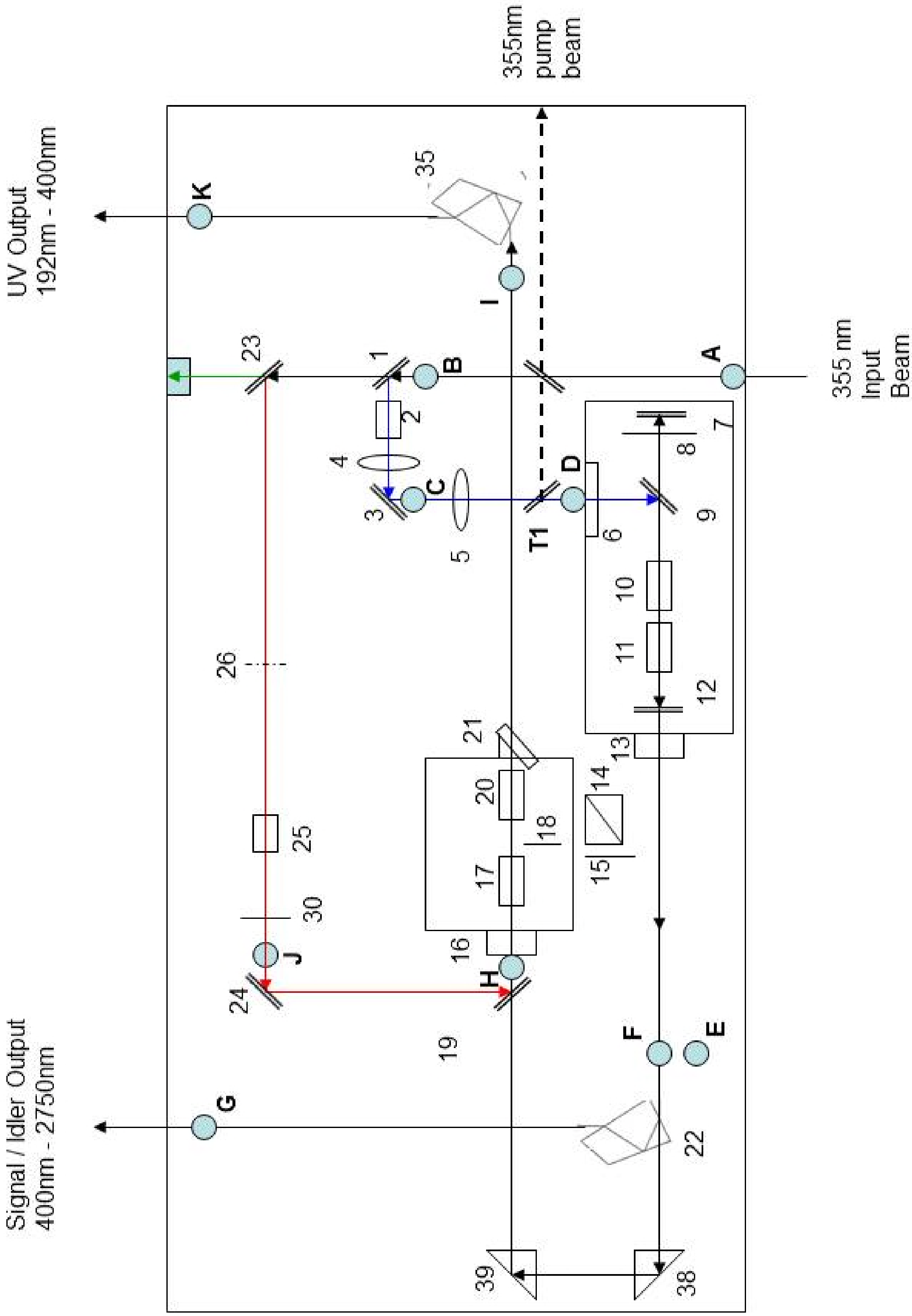


Figure 3.1.1
Horizon OPO optical layout with alignment gig locations



B. Complete alignment

(This should only be required if your system has been moved to another location; if there is a change of pump laser, or if significant power loss or instability has been observed.)

Pump laser optimization

Characteristics of the pump laser are critical to the overall performance of the Horizon OPO. It is important to take time to optimize the pump beam for energy and beam quality before aligning through the OPO. Refer to the pump laser beam burns that were taken at initial installation as a reference.

C. Alignment tools

Assemble these items before adjusting or aligning Horizon OPO optics:

- Metric Allen wrench set
- Red phosphor paper, business card, burn paper and plastic bags
- 2 pinhole jigs (998-0182) with red paper around one side of their pinholes; pinholes are 43 mm above the bench. (provided with your Horizon)
- Power meter with 10-30 watt power absorbing head
- Fast photodiode and oscilloscope (350 MHz bandwidth)

Complete pump beam alignment

1) Approximately level the pump laser by adjusting its feet and then make sure that the bolts securing it to the bench are tight. Placing a round bubble level on its optical bench will assist in leveling.

2) Close the pump laser shutter for your safety and warm up the pump laser for approximately 30 minutes in flash mode. This allows the laser head and harmonic crystal housings to temperature stabilize. If seeding, the seeder unit should be switched on as this takes a little longer to come up to thermal equilibrium – approximately 30-45 minutes.

3) When ready for alignment, activate the Q-switch and proceed to delay this until the energy output is as low as 1mJ at 355nm. This gives a safe value for initial alignment; once confident in the alignment, the Q-switch can be adjusted back to the normal operating value for an optimal 355nm output reading.

4) Place the Horizon OPO in front of your pump laser and position so that the 355nm input beam will be centered on the input port.

5) Open the Horizon OPO lid and install the Interlock Defeat Bracket (655-1087) that defeats the safety interlock.

Warning!

Wear goggles and observe all safety precautions when operating the Horizon OPO with its cover removed.

6) Verify that the Horizon's 12VDC power supply is connected to the Horizon, plugged into AC power and has been running for a minimum of 10 minutes to allow the OPO and UV crystal heaters to stabilize.

7) Place the Pinhole at position A in your Horizon. This will be the alignment target for your input Nd:YAG laser. The 355nm beam should hit the center of the 43-mm pinhole at location A and B. If not, either the Horizon or YAG laser legs can be adjusted. The beam will pass through pinhole A and should also pass through the 43-mm pinhole at location B.

Complete Horizon OPO pump beam alignment

1) Bolt both the pump laser and Horizon securely to the table.

2) Again check pump alignment through pinhole at A and B.

3) Relocate the alignment jig to location C. Adjust mirror 1 to center the pump beam on the pinhole.

4) Place second alignment jig in location D. Adjust mirror 3 to center the pump beam on the pinhole.

5) Repeat steps 3 and 4 a few times until the pump beam is centered on both pinhole test jigs C and D simultaneously.

6) Remove both pinhole test jigs.

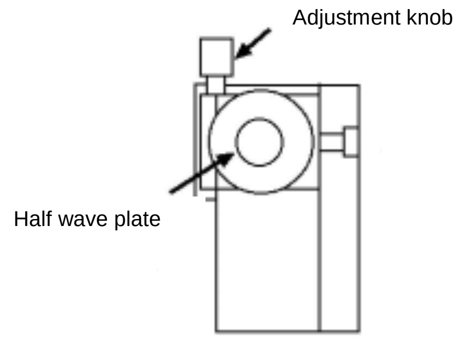
7) Deactivate the Q-switch and close pump laser shutter.

8) Install tool T1 between location C and D per figure 3-1. Remove the plug from the side exit of the Horizon to allow the 355nm beam to pass through.

9) Open pump laser shutter and activate the Q-switch. At this point you are still running at 1mj OF 255nm. Ensure that the beam passes through the side exit cleanly.

10) Turn on laser in a full-power mode and optimize harmonic crystals so that its 355 nm output is running at the specified pump level for your OPO (see the list of pump laser specs on page 21). If you have the optional attenuator, ensure that it has been set at maximum throughput when measuring the pump level into the OPO.

Figure 3.2. Horizon OPO Oscillator Attenuator



To change the attenuation, rotate the half wave plate with the adjustment knob.

A 45° rotation changes the attenuation from 5% to 98%

11) The pump beam collimation should be checked from T1 to a distance of approximately 2.5 meters from the output port of the Horizon at full energy. If the beam is not collimated, please contact a Continuum Service Engineer for assistance and troubleshooting.

12) Take full power burns of the 355nm pump beam at the side exit of the Horizon. Compare these with burns taken at installation. A small degradation in beam shape or energy may be an indication that a pump laser adjustment is necessary. Please refer to the pump laser manual for more directions to optimize the pump beam quality. Refer to page 21 for the energy requirements for your particular pump laser.

13) Again reduce the 355nm pump level to 1mJ continue the alignment procedure.

14) The pump beam should still be on the 43mm pinhole at position D. Adjust mirror 3 if necessary.

15) Place 43-mm pinhole jig in location F. Open the oscillator top housing cover to access mirror 9. Adjust mirror 9 vertical or horizontal top adjustments to center the beam on this pinhole.

16) Place 43-mm pinhole jig in location A. Adjust output optic 12 (vertical and horizontal) so that the back reflected pump beam goes through the pinhole.

17) Repeat steps 14 – 15 if necessary.

18) After ensuring the beam characteristics of the Nd:YAG pump laser at 355nm – no hot spots, good symmetry and energy – and the pump beam position is correct for the Horizon then you are ready to align the OPO

Complete Horizon OPO alignment

1) Open the Horizon GUI software.

2) **GoTo** 508nm using the GUI.

3) Turn pump laser to full power.

4) Open ***Tools*** from the menu bar and select ***Crystal Calibration*** from the GUI.

5) Adjust Crystal 2 motor and select step size of 10 to make adjustments to maximize the output power.

6) Remove the oscillator cover to access optic 7. Finely adjust the horizontal and vertical controls for optic 7 to maximize the output energy.

7) Repeat steps 5 – 6 until output power is maximized.

8) Follow the following section - ***Horizon OPO Oscillator Calibration -*** to save data to the calibration file.

D. Horizon OPO oscillator calibration

1) To calibrate Horizon to specified precision a spectral device with 0.1 nm resolution is required.

2) Open ***Tools*** from menu bar in the GUI and select ***Crystal Calibration***. Goto 508nm and monitor the output power. Adjust crystal 2 position in incremental steps of 10 to maximize the output energy.

3) Place the 43-mm pinhole jig in location G. Change Prism 2 step to center the output beam on the pinhole.

4) Remove the 43-mm pinhole jig in location G. Read wavemeter reading. Enter this number into the wavelength input number in the calibration window.

5) DO NOT Click **Set** (very important) Click ***<u>Add</u>*** and a window will give you the option to ***Save AS.***

Warning!

<u>Do not</u> click ***"Set"*** as this will write over the calibration files set up in the factory.

6) Select ***Save As*** and the positions of crystal 1, 2, and prism 2 will be save to the calibration file. *Note* – there is a second request for ***Save As*** embedded in the software to ensure these changes.

7) Your OPO is now calibrated. This new value will automatically offset the original calibration and be ready to run.

E. Horizon OPO UV generation

Phase matching UV crystals

The Horizon OPO output wavelength is defined by the OPO crystal orientation. The crystals and Pellin Broca are driven by high precision motors – these positions are calibrated to ensure accurate and repeatable performance. The crystals and Pellin Broca are calibrated to work in sink with the Horizon's oscillator. This ensures exact phase matching as the Horizon is scanned to maximize the output at all wavelengths.

Warning!

Wear goggles and observe all safety precautions.

The UV option extends your Horizon OPO's performance into the UV and deep UV. A complete tuning range from 192-2750nm can be simply accessed. The UV package can provide more than 20 mJ of UV energy (depending on the pump laser) during scans exceeding 200 nm in length, while its rotating Pellin-Broca output prism removes residual signal and idler and maintains constant output beam pointing. Highly accurate calibration tables and stepper motors independently control crystal rotation, obviating any need for feedback to phase match the crystals. Calibration control of crystal rotation simplifies and speeds up scanning while reproducibly optimizing phase matching at all wavelengths.

F. Horizon UV alignment

Caution!

Perform these alignment steps at Horizon OPO energy; it is convenient to use a visible wavelength.

1) Increase the Q-switch or Amplifier delay in the PL DLS 9000 so that the residual IR power after mirror 23 is sufficiently reduced for alignment.

2) Place 43-mm pinhole jig in location J. Adjust mirror 23 so that the IR beam is centered on the pinhole.

3) Place 43-mm pinhole jigs in location H and I. Adjust mirrors 24 to make the IR beam centered on the pinhole at location H

4) Adjust mirror 10 to make the small beam after the pinhole at H goes through the pinhole at location I.

5) Repeat steps 5 and 6 a few times until the IR beam is centered on the pinhole at H, and the small beam after the pinhole at H goes through the pinhole at location I.

6) Block the IR beam with the beam block provided. Increase the pump energy until there is signal output from the OPO.

7) Adjust prisms 38 to make the signal beam centered on the pinhole at H.

8) Adjust prism 39 to make the small beam after the pinhole at H goes through the pinhole at I.

9) Repeat steps 7 and 8 a few times until the signal beam is centered on the pinhole at H, and the small beam after the pinhole at H goes through the pinhole at location I.

G. Horizon UV calibration

The Horizon's UV housing contains two crystals to address the entire 192-400nm tuning range. Calibration is selected at different ranges to ensure optimal output throughout. As expected this harmonic module tracks with the oscillator output. The crystals are calibrated to give maximum energy output at each wavelength. The optimal phase match angle is determined and at that point the exact motor position recorder and stored in the GUI software. The calibration values should not need adjustment as all values have been preset at the factory. However, if there is a specific wavelength where optimal output is required (shifted from the peak), the oscillator calibration curve can be tweaked slightly and the UV adjusted accordingly. Please contact our Service support line for assistance if needed.

292nm – 400nm calibration

1) Set the GUI "GoTo" menu to go to 340nm. From the Tools menu bar open the ***Crystal Calibration*** window. You will notice that there are six motors listed, four are for controlling the OPO and UV crystals, and two for controlling the Pellin Brocas. When observing the UV output power on a power meter fine adjustments can be made to the crystal position to maximize the output energy. 10 step increments can be taken for optimizing, smaller steps should not be necessary.

2) Once optimized, move the 43-mm pinhole jig to location K (by the UV output port). Adjust prism 1 step to center beam on pinhole, Remove Pinhole.

3) Click ***Add*** and then ***Save As*** in the calibration window.

4) ***Save As*** will save the new positions for crystal 4 and Prism 2 into the calibration files. If you use ***Save***, this will overwrite the current calibration file.

Caution!

DO NOT SELECT SAVE.

234nm – 292nm calibration

1) Use GUI ***GoTo*** menu to go to 260nm. Open the ***Crystal Calibration*** window again. Follow the same procedure outlined above. In this case, only motor 4 is adjusted. It is no longer necessary to adjust Prism 1 as it has already been calibrated.

208nm – 234nm calibration

1) Use GUI ***GoTo*** menu to go to 220nm. Open the ***Crystal Calibration*** window. Using the same procedure as above, now crystal 3 can be adjusted for the maximum output. Again, it is no longer necessary to adjust Prism 1 as it has already been calibrated.

193nm – 208nm calibration

1) Use GUI ***GoTo*** menu to go to 200nm. Open the ***Crystal Calibration*** window.

2) Change crystal 3 in steps to visually maximize the beam next to the output beam. (This beam is blocked by a beam block. It has to be checked inside the laser close to the output port.)

3) Change crystal 4 step to maximize mixing output energy.

4) Change crystal 3 and crystal 4 steps until maximum mixing output is obtained.

5) Click ***Add*** then ***Save As***.

6) Click ***Save As*** to store the new positioning for crystals 3 and 4 in the calibration files.

Warning!

DO press ***Save*** as this will overwrite the current calibration file.

Chapter IV Horizon OPO System Description

The Horizon OPO is an optical parametric oscillator system generating mid-band radiation in the visible and near infrared spectral region. Highly efficient single stage oscillator produces coherent light between 400 and 2750 nanometers from a 355 nm Nd:YAG pump laser beam. **Figure 3.1** describes the Horizon OPO optical configuration. An advanced computer control system maintains optimum crystal performance (phase matching) while scanning over a wide wavelength range.

The Horizon OPO is designed to be pumped by either a Surelite or a Powerlite OPO pump laser. This efficient OPO includes full digital motor control of crystals and wavelength separation for optimal output at all times. The Horizon also includes a UV module to give complete wavelength coverage from 192-2750nm. The signal/idler exits from one port and the UV exits from a second port.

Optical parametric conversion

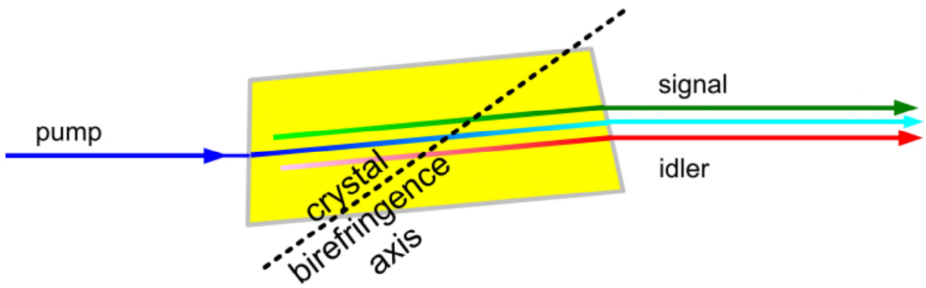
The Horizon OPO high performance optical parametric converter provides solid-state reliability, high spectral brightness and continuous tunability in the visible and near infrared spectral region. Pumped by 355 nm light from an Nd:YAG laser. Horizon OPO parametric converters feature excellent beam quality, high peak and average power and mid-bandwidth across a very broad tuning range

The optical parametric process

An optical parametric process is a three-photon interaction, in which one photon (the pump photon) splits into a pair of less energetic ones. The resulting two photons will not, in general, have the same energies. The higher energy photon produced is referred to as the signal, while the lower energy photon is called the idler (see Figure 4.1).

This energy conversion process may be thought of as the reverse of sum frequency generation, where two photons combine to form a third, more energetic photon.

Figure 4.1. An optical parametric process.



OPO processes must satisfy conservation rules

As is in other coherent interactions, energy and the momentum vector must be conserved during an optical parametric process. That is, an optical parametric process occurs when energy is conserved,

$w_p = w_s + w_i$ where w is the photon energy,

and when momentum is conserved,

$\mathbf{k}_p = \mathbf{k}_s + \mathbf{k}_i$ where **k** is the photon's momentum vector.

The most common method to achieve the simultaneous conservation of energy and momentum takes advantage of the birefringence of a transparent nonlinear crystal such as BBO (beta-barium borate). Such materials have different indices of refraction for different polarizations of light. One index varies with changes in propagation direction with respect to the crystal angle.

Energy conservation is satisfied when the sum of signal and idler frequencies equals the pump frequency. However, linear momentum will not, in general, be conserved, because the two resultant photons have different frequencies than the pump, and velocity is dependent on frequency. But when the pump, idler and/or signal photons have polarizations orthogonal to the pump photon, there can exist angles at which the crystal's frequency-dependent refractive index changes the pump, idler and signal velocities so as to conserve linear momentum. Since the pump and signal/idler pair polarizations conserve angular momentum, the crystal then fulfills all conservation conditions necessary for parametric conversion

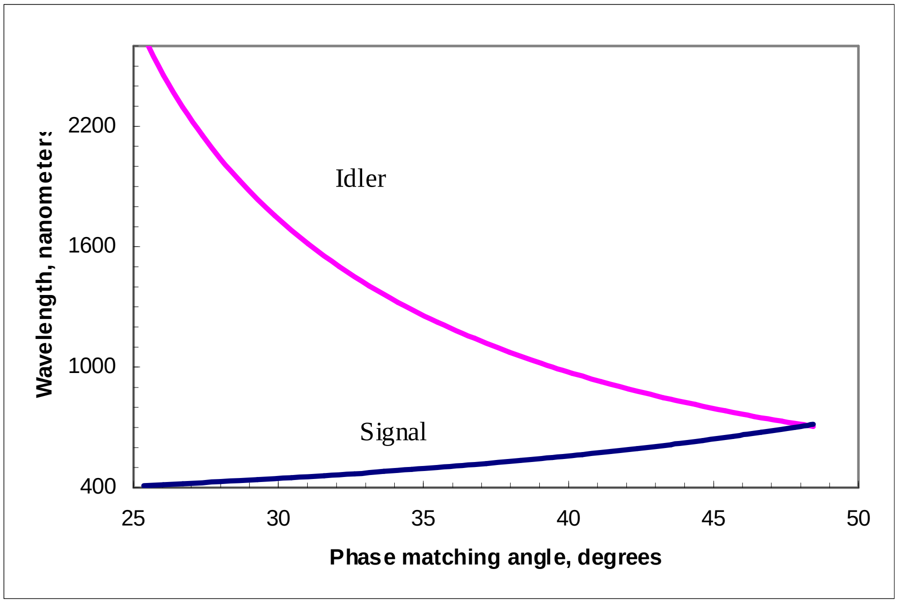
Phase matching selects OPO output frequencies

The angle the crystal axis makes with respect to the pump beam determines both the signal and idler frequencies at which momentum is conserved. The angle tuning of birefringent materials to produce a particular signal/idler pair is called phase matching. Both the angular range over which it phase matches and its absorption characteristics determine a crystal's tuning range; see **Figure 4.2**.

The Horizon OPO produces efficient optical parametric frequency conversion over a wide spectral range. A sample spectrum of the Horizon OPO output from 400 nm to 2750 nm is shown in Figure 4.4 later in this Chapter.

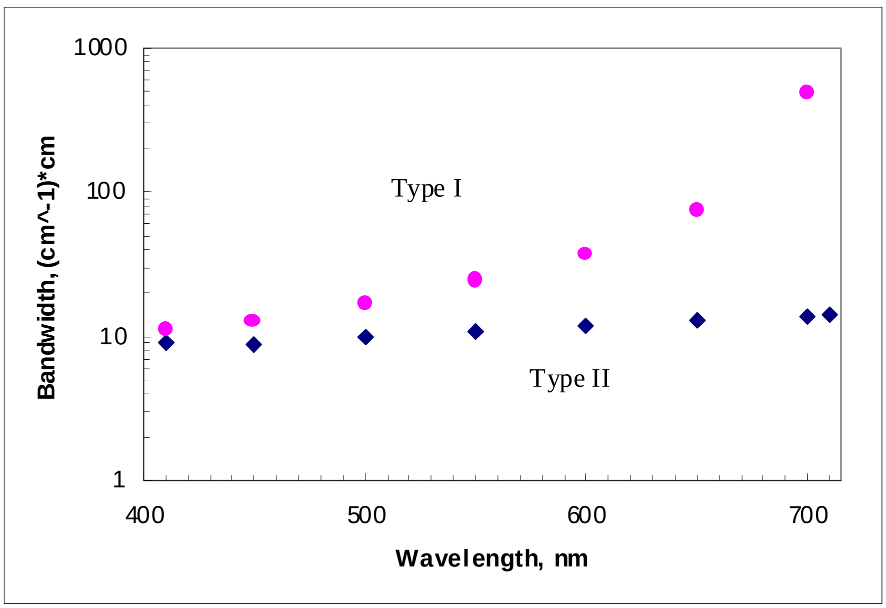
The Horizon OPO utilizes Type II phase matching in the BBO crystal. It means that the extraordinary pump wave (with horizontal polarization) interacts with ordinary (vertical) signal and extraordinary (horizontal) idler waves. This interaction has inherently narrower acceptance bandwidth, which produces narrower output linewidths (compared to Type I interaction).

Figure 4.2 Horizon OPO output varies from 400nm to 2750nm



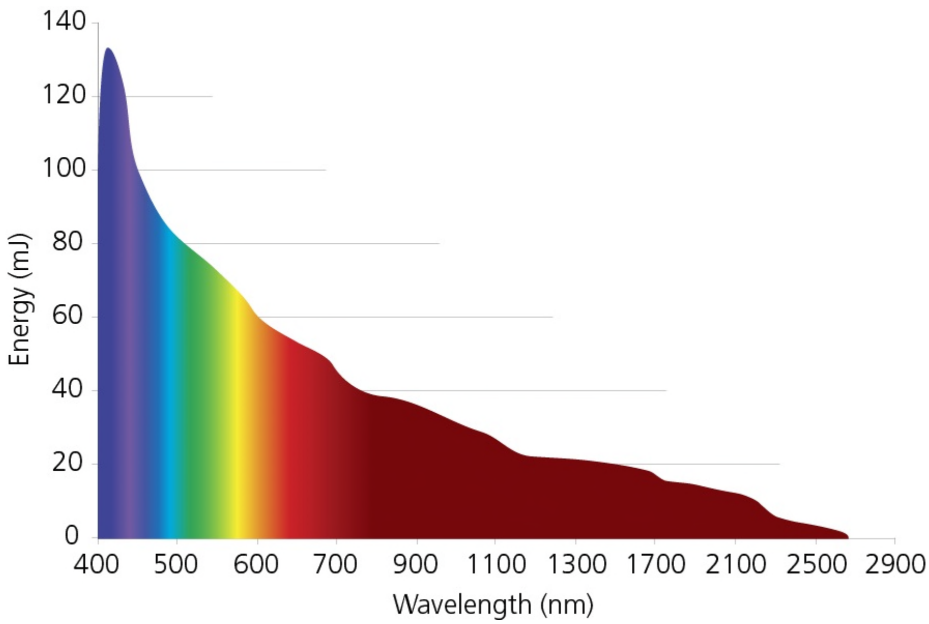
Seeding improves Horizon OPO linewidth & efficiency

Figure 4.3. Acceptance bandwidth in Type 1 and Type 2 BBO OPO



A pump beam, traveling through an appropriate crystal at the angle satisfying conservation laws, normally undergoes parametric conversions at low efficiency; most of the injected beam emerges unchanged. The longer the interaction length in the crystal is, the higher the conversion efficiency is. Horizon OPO uses two strategies to increase conversion efficiencies.

Figure 4.4
Horizon performance with PL 8000 pump

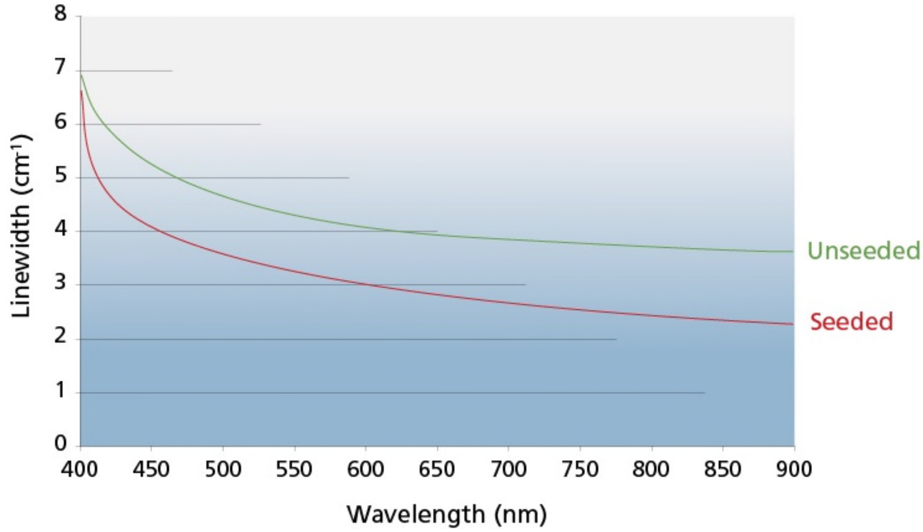


The first strategy is to use the output mirror of the Horizon OPO oscillator to reflect most of the undepleted pump pulse energy back into the nonlinear crystal, thus doubling the interaction length between the pump and the generated waves. Additionally, the Horizon OPO oscillator was designed to be as short as possible, to increase the number of round-trips of the oscillating wave during the duration of the pump pulse. This also increases the interaction length and efficiency of parametric conversion.

The second advantage is a longer interaction length in the nonlinear crystal. The linewidth of the generated signal and idler waves is shorter when the crystal length is longer. The linewidth of the OPO output waves is proportional to the acceptance bandwidth of the nonlinear crystal divided by the square root of the number of round-trips in the

Figure 4.5. Horizon OPO typical output linewidth

Pump laser: Powerlite 8000 (290mJ at 355nm)



OPO oscillator (interaction length in other words). The Horizon OPO cavity allows to generate less than 3-7-cm^{-1} output bandwidth in the whole tuning range, using as a pump simple Surelite lasers or unseeded Powerlite lasers.

When the Horizon OPO is pumped by one of the seeded Powerlite lasers, the output linewidth can be as narrow as 2 cm^{-1}. There are two reasons for linewidth narrowing. Firstly, laser linewidth changes from ~1.5 cm^{-1} in the unseeded mode of operation, to <0.01 cm^{-1} in the seeded mode. The output linewidth depends on a superposition of the crystal's acceptance bandwidth and the pump linewidth. Narrower pump linewidth directly decreases the output linewidth. Secondly, the pump pulse from the unseeded laser has strong amplitude modulation due to competition of different spectral components, whereas the pulse from the seeded laser has a smooth shape with approximately the same envelope duration. The energy of the unseeded pulse can be the same as that of the seeded pulse, but the higher intensities of the multiple peaks cause broader output linewidth.

Computer system

The Horizon OPO produces coherent light over a wide wavelength range, and its oscillator crystal angle selects this wavelength. Horizon OPO software maintains the phase match angle of each of the crystals, rotating them in concert when changing wavelength.

User interface

Because the computer is sometimes out of reach, a mouse on a long cable controls all software; an on-screen numeric keypad even permits the mouse to enter numerical values. The user interface is a single window which displays the current status and controls wavelength, scanning and direct crystal movement. Additional windows appear inside this main window during file management and calibration operations. Please see the GUI manual for the software operator's manual.

Automatic phase matching

Horizon OPO software uses its crystal calibration tables to keep all crystal angles and Pellin Broca optimized during all wavelength changes. There are no active feedback loops. The exact motor position for each crystal throughout the tuning range is captured and stored in the software.

Wavelength control

The angular position of the oscillator BBO crystals selects both the wavelength and one crystal also acts as a compensator to ensure that there is no beam walk-off as the system is scanned. We know that this feature is critical for any experimental set up.

Wavelength scanning

Besides manually moving individual crystals to phase match, there are a number of ways to move crystals simultaneously and change wavelength while maintaining optimum crystal angle tuning. In the Horizon, all motions are linear in units of nm/sec, and are controlled by an intuitive GUI interface. GoTo, Internal, External Scan and Burst mode options are all available.

Scan motions:

***GOTO*:** Moves to a selected wavelength quickly – this is not a scan to mode. The advantage of digital motor control is the capability to "Jump To" a new wavelength rather than "Scan To". This speeds up the overall process.

Speed: Moves in the chosen direction at a chosen speed

Scan: Three types; each begins and ends at a selected wave length:

- Normal: scans, returns, and repeats as desired.
- Internal: scans, waits for a fixed time and then continues to scan again.
- External mode – drive your scan externally.

Backlash compensation

Now with the use of high precision digital motor controllers, backlash is no longer a concern. Each step can be counted and registered for instant recall. The result is linear scanning and repeatable performance.

Park feature

The GUI software has been programmed to ensure that all crystals are parked safely at the end of each day. Simply close the operating platform in a standard mode and the safety feature will be activated.

Electronics

Temperature controllers

Individual temperature controllers for each crystal ensure reproducible behavior. Each controller is a small PCB mounted inside the sealed crystal oven directly below its crystal mount, with its thermal sensor and heater in contact with the base of the mount.

Crystal controllers

A controller card in the Horizon sends signals which move each of the crystal motors and wavelength separation Pellin Broca prisms directly. These motors require no other control electronics.

Glossary

Common terms used by Continuum in describing parametric converters:

Birefringence	An optical property of certain crystals. Such materials have different indices of refraction for different polarizations of light, and one index changes with changes in

	propagation direction through the crystal.
BBO	β- barium borate, a non-linear crystal with good transmission qualities and a high damage threshold, used in the Horizon OPO.
Idler wave	In the three-wave process, the idler is the less energetic resultant wave.
OPO	(Optical Parametric Oscillator): A resonator constructed with a non-linear gain medium. A generated wave (in this case, the signal) is fed back by the resonator to the non-linear crystal causing further generation, or amplification, of the wave. An OPO is analogous to a laser resonator, except that its gain is a coherent process, as opposed to a laser's population process. This manual also uses this abbreviation to refer to the first stage of the Horizon OPO, an oscillator.
Phase matching	The process of conserving linear momentum when down-converting a pump wave to a signal wave and an idler wave. The Horizon OPO accomplishes phase matching by changing the crystal angle with respect to the pump beam; this is called angle tuning. Changing the crystal under computer control maintains the optimum angle at all wavelengths.
Pump wave	In the three-wave process, the pump is the input wave; it is the most energetic of the three.
Signal wave	In the three-wave process, the signal is the intermediate energy wave; it is the more energetic output wave.

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Chapter V Routine Maintenance

Maintaining optics

Periodically inspecting the surfaces of all crystals and optical components for dust, discoloration, and damage, and cleaning or replacing them if necessary will promote good system performance and longer lifetime. Be sure to replace the crystal oven lids and large plastic cover when finished.

Tools and supplies required

Keep these tools and supplies available for examining, cleaning and replacing optics:

- finger cots and lint-free fine cotton gloves
- high-intensity flashlight and inspection mirror
- dry, reagent grade (or better) methanol and/or acetone
- eyedropper, lint-free lens tissues and lint-free cotton swabs
- source of clean, dry gas under pressure (not Freon-based)
- metric Allen wrenches
- hemostat.

Inspecting and dusting optics

Dust on optical surfaces absorbs laser light and the hot spot will burn the surface. Regular inspection and cleaning with compressed gas can prevent damage. Follow these steps to dust the optics and determine whether an optical surface needs cleaning or a component requires replacement.

1) Turn the pump laser off and remove the Horizon OPO black anodized aluminum lid.

2) Systematically examine the surfaces of each lens, crystal, grating and oven window for dust, fingerprints, discoloration, and burn marks in the surface coating. Use a bright flashlight to light the surfaces and eliminate reflections, and an inspection mirror where needed. Do not open the crystal ovens for inspection, but examine the exterior of their windows.

3) Clear dust and loose debris present on the face of any optic with clean, dry compressed gas (dry nitrogen is preferred).

4) Note which components need cleaning and which need replacing. Don’t clean any component unless compressed gas is ineffective.

Cleaning optics

1) Wash hands and put on gloves or finger cots.

2) Place one sheet of lens tissue over optic to be cleaned (use a hemostat if necessary). Using an eyedropper, place a few drops of reagent grade methanol on top of the tissue.

3) Drag the lens tissue across the optic once only. If a visible solvent residue remains on the optic, repeat using less solvent and a new tissue until no residue remains.

4) For hard to reach optics, wrap lens tissue on a cotton swab. Place a few drops of solvent on the tissue; shake off any excess.

5) Swab the optic gently with this applicator. Repeat these steps with a drier swab to remove remaining residue.

Handling BBO crystals

BBO crystals are hygroscopic and must be scrupulously protected from water. They are also extremely sensitive to burning caused by light striking dust on their faces. Before handling BBO crystals, remember:

- Use dust-free anti-static finger cots, not gloves, to handle crystals.

- Also wear an anti-static wrist strap when handling crystals.

- Never touch a crystal’s end face, even with finger cots

- Clean crystals only with dry, oil-free, filtered nitrogen or a dust-free Q-tip.

- Keep the nitrogen nozzle at least six inches from all crystal faces

- Never use a solvent or lens tissue on a crystal.

- Keep the crystals out of room air as much as possible.

- Keep BBO enclosures dust free, because BBO develops a static charge when pumped, attracting dust.

- Keep burn paper and white business cards in plastic bags and at least 6” from all crystals, even at low power; dust and dyes will reach the crystal faces and cause burning.

- BBO is brittle; over-clamping or careless handling can cause chipping.

- Over-clamping will strain the BBO and cause unwanted birefringence, degrading the parametric process.

- Defects in a crystal coating will burn almost immediately, so inspect faces for spots, streaks and ‘fish eyes’ before use.

Cleaning BBO crystals

Never touch a crystal with tissue or a solvent. Blow dust off with a stream of clean, dry gas, and avoid touching it with anything else. Follow all cautions in the above section on handling BBO crystals. If nitrogen does not work, please call Continuum for assistance.

Replacing an optic

Some optics have wedges and must be replaced with a new optic in the same orientation, and the half wave plates also have orientations which must be aligned.

Note:

Replacing an optic requires realignment; do not attempt this unless you are familiar with the alignment procedure. Please check with Continuum before replacing an optic.

Horizon OPO optics have a three-point mount, and are secured with a 2.0 mm set screw. To replace an optic:

1) Note the position of the optic in the mount and loosen its set screw, located at the top of the mount. Push the optic out of the mount with a cotton swab.

2) Wearing gloves, place the new component into the mount and push it into the previous position. Make sure that the orientation is correct. Tighten the set screw.

Note:

Over-tightening the set screw may damage the optic.

3) Check the optic for dirt or dust before remounting.

4) Realign the optic, referring to the appropriate procedure in Chapter III.

Replacing the oscillator crystal

Damage to an oscillator crystal face will reduce oscillator output, and it is not possible to reposition the crystal to expose a fresh surface. If replacing the crystal, return the damaged one to Continuum for repolishing.

Note:

Replacing the oscillator crystal requires recalibrating. Continuum will advise you on receipt of the crystal whether we must reinstall and align it. If not familiar with recalibration, contact Continuum service before removing the crystal.

1) Change the OPO wavelength to 508 nm. At this wavelength the OPO crystal faces are almost perpendicular to the pump beam.

2) With the YAG pump beam blocked, remove the six M5 bolts holding down the oscillator cover. Remove the cover, and replace the desiccant bags with fresh ones.

Caution!

Read and observe the section on Handling BBO crystals earlier in this Chapter. The slightest amount of moisture or dirt will damage the crystal. Always wear finger cots when handling the crystals.

3) The OPO crystal comes mounted into its holder with the heater board installed. The holder is shipped in the dry box and has a protective plastic cover.

4) Remove the top lid from the dry box and unscrew two screws retaining the crystal holder on the shaft. Remove the OPO crystal holder from the dry box.

5) Unscrew two screws retaining the OPO crystal holder on the shaft in the OPO housing. Disconnect the two wires from the heater board. Now remove the crystal holder with the damaged OPO crystal from the housing.

6) Install the new crystal holder with the new crystal on a shaft. The new crystal should be mounted to be parallel with the other. Tighten both screws retaining the crystal holder on the shaft. Remove the protective cover from the crystal holder. Clean the new crystal with dry nitrogen.

7) Reconnect the two wires from the crystal heater boards – pay attention to the “+” and “-” polarities.

8) Re-install the OPO housing cover.

9) Alignment procedure is required when the OPO crystal is replaced. (See Chapter III)

Calibration

The Horizon OPO calibration files control all the motorized optics in your system; two crystals in the oscillator, the signal/idler Pellin Broca, two crystals in the UV package and an additional Pellin Broca for the UV wavelength separation. The crystal calibration files have all been set in the factory and the correct phase match angle for all wavelengths for each crystal optimized. At installation the Continuum Service Engineer will ensure that the calibration files are still at the maximum readouts. If something moves in shipment, a fine adjustment can be made and that information saved.

Crystal calibration

Each crystal has a calibration table, which associates each Horizon OPO wavelength with the crystal motor position at which the crystal is optimally phase matched. The calibration tables can be access from eth GUI pull down menu. You will see that there is an entry for each nm over the entire wavelength range. During wavelength changes, software interpolates between the two closest wavelengths to determine the crystal motor position which maintains phase matching.

The calibration procedure rotates the crystal, which allows you to phase match it after each step, and then writes the motor position and wavelength into a line of the table. After this process, you may observe and edit the table, and write it into a calibration file on the hard disk when satisfied. We advise you talking to a service Engineer before making any modifications.

These calibrations do sometimes change, especially if the pump laser is moved. It is also desirable to recalibrate when a crystal regularly becomes detuned after long wavelength scans.

Multiple calibration files can be stored and recalled for use; see the File menu commands. Calibration files can be made new or remade by editing an existing table.

The OPO crystal is calibrated using the wavelength measuring equipment and the UV crystals are calibrated by optimizing phase-matching for the given signal wavelength. To calibrate the Horizon to a specified precision, a spectral device with 0.1nm resolution is required. The Burleigh Wavemeter operating with etalon A (low resolution) provides sufficient precision for the Horizon calibration.

The output wavelength depends only on the OPO crystal orientation in the OPO oscillator resonator.

Chapter VI Troubleshooting

General troubleshooting guide

Problem	Possible Solutions
Oscillator will not resonate	1) Inspect the oscillator for obvious problems or blockages. 2) Check the pump seeding, beam shape, power and output pointing. 3) Check that the attenuator (BA) transmits enough energy into the OPO. 4) Check the pump beam alignment at positions A and B. 5) Realign the oscillator from scratch (Chapter III).
OPO stops oscillating as the beam alignment at positions A and B. changes	1) Realign the oscillator from scratch (pages 3-7 to 3-12). 2) Recalibrate the OPO crystal (Chapter V).
OPO output is erratic and/or low in energy	1) Check the injection-seeding of the pump, as well as its pulse-to-pulse energy, and its beam alignment 2) Ensure that the optional pump beam attenuator transmits enough energy into the OPO. 3) Check crystal 1 and 2 calibrations. 4) Examine the cavity for dust and/or damage. Clean or replace affected optics (Chapter V)
Wavelength scanning not smooth and continuous	1) Check crystal 1 & 2 calibration
Pump clips or damages a variable attenuator	1) Re-align the pump beam (Chapter III) 2) Open and dust the attenuator and check all pump optics.
Snapping sound heard	1) Gross misalignment of the pump beam into the crystals is likely. Stop operation immediately and inspect for damage to crystals and optics. Realign pump (Chapter III); clean or replace optics. 2) If no damage is found and the noise occurs around the oscillator output, water vapor may be the cause. Usually the noise appears ~1.4 µm.

Repeated optics and/or crystal damage	1) The pump laser needs routine or major servicing. Check your pump laser service manual for routine service instructions. Take burn patterns looking for abnormalities or hot spots. 2) The pump intensity is too high due to incorrect operation. Check and correct as required. 3) Dust is accumulating on the optics and crystal. Check for sources of dirt. Check for reflections striking surfaces and causing ablation. Clean the crystal and remove the cause of the dust (Chapter V). 4) The pump beam is striking a crystal edge as the crystal is rotated to large angles or reflecting off an internal surface. Re-center the pump beams and recheck its alignment (Chapter III). 5) Damaged pump optics or pump beam hot spots are 'avalanching' into the crystals; take pump beam burns. 6) Check the crystal for bubbles and/or volume damage.
Low energy in doubling output while scanning	1) Check OPO signal output energy 2) If OPO output energy is normal, recalibrate the crystals 3 & 4.
Poor energy stability	1) Check the stability of the injection-seeding of the pump. Check the output energy stability of the pump. Perform normal pump laser maintenance. 2) Check the stability of the OPO output. 3) Check that the crystal heaters are on.
Phase matching drifts over minutes	1) Check that the crystal heaters are on. 2) Check the crystal heaters by opening the oven and placing a thermometer on the crystal stage.
Output wavelength is not exact	Recalibrate crystals 1 & 2.

Chapter VII Horizon

Express warranty

Unless otherwise specified, all mechanical, electrical and electro-optical components made by Continuum are warranted to be free from defects in materials and workmanship for 1 year. Optics and crystals have a 90-day warranty. These warranty periods apply only to systems delivered to the United States and Canada. This warranty is in lieu of all other warranties, express or implied. No warranties of Merchantability or fitness or any other remedies are available.

Limitations

This warranty's remedy is limited to repair or replacement of the defective materials, subject to these conditions:

This warranty does not apply to materials which Continuum decides have been damaged by abuse, misuse, mishandling, accidental alteration, negligence, improper operation or other conditions not under Continuum's control.

This warranty does not apply if the original identification markings have been removed, defaced or altered, or if modifications or substitutions have been made without Continuum's prior consent.

This warranty does not apply if the customer's account is delinquent.

For warranty periods of products sold outside the United States and Canada, please contact your local representative or sales office.

Returns, adjustments and service

To request any warranty or other repair or service to any Continuum product:

Obtain a return authorization number from Continuum's Service Department.

Package the product properly using the original shipping container. Pre-pay freight and full-value insurance, and assume all risks of loss, damage, or delay in shipment, and return it to the authorized address.

Upon arrival, Continuum will examine the product to determine the cause of failure and warranty status. Continuum is not obligated to perform a warranty repair if shipping damage obscures whether any warranty defect existed.

The warranty period on a repaired or replaced product is the balance of the warranty period of the original product. If the product is not under warranty, Continuum will advise the customer of repair charges and require a written purchase order for repairs before work begins.

Continuum service centers

For most current information regarding Continuum service center in your area, please visit: **http://www.continuumlasers.com/contact_info/**

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Operator Notes